

1. 다음 급수의 수렴 및 발산을 판정하면?

(1) $\sum_{n=2}^{\infty} \frac{1}{n^2 \ln n}$

(2) $\sum_{n=2}^{\infty} \frac{1}{n \log n}$

(3) $\sum_{n=2}^{\infty} \frac{1}{n (\ln n)^3}$

(4) $\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$

(5) $\sum_{n=2}^{\infty} \frac{\ln n}{n(n+1)(n+2)}$

(6) $\sum_{n=2}^{\infty} \frac{\log n}{2n^3 - 1}$

(7) $\sum_{n=2}^{\infty} \frac{n^2 + n + 1}{\ln n (n^3 + n + 1)}$

(8) $\sum_{n=2}^{\infty} \frac{1}{n \ln(n+1)}$

(9) $\sum_{n=2}^{\infty} \frac{1}{\ln n}$

(10) $\sum_{n=2}^{\infty} \frac{1}{\ln n \ln(n+1)}$

(11) $\sum_{n=2}^{\infty} \frac{1}{(\ln n)^5}$

(12) $\sum_{n=2}^{\infty} \frac{n^2}{3^n}$

(13) $\sum_{k=1}^{\infty} \frac{2^k}{k^3}$

(14) $\sum_{n=1}^{\infty} \frac{3^n}{n(n+1)}$

(15) $\sum_{n=2}^{\infty} \frac{(1.01)^n}{n^{100}}$

(16) $\sum_{n=2}^{\infty} \frac{(n+1)^5}{2^n}$

(17) $\sum_{n=1}^{\infty} \frac{3n^2 + 1}{e^n}$

(18) $\sum_{n=1}^{\infty} \frac{5^n}{100^n}$

(19) $\sum_{n=1}^{\infty} \frac{10^n}{8^n + 9^n}$

(20) $\sum_{n=2}^{\infty} \frac{2^n}{n^3 + 2^n}$

(21) $\sum_{n=2}^{\infty} \frac{\ln n}{\sqrt{n} e^n}$

(22) $\sum_{n=2}^{\infty} \frac{5^{2n}}{\sqrt{n} 9^n}$

(23) $\sum_{n=1}^{\infty} \frac{n^3}{n!}$

(24) $\sum_{n=1}^{\infty} \frac{5^n}{n!}$

(25) $\sum_{n=1}^{\infty} \frac{n!}{e^n}$

(26) $\sum_{n=1}^{\infty} \frac{n!}{\ln n (n^3 + n + 1)}$

(27) $\sum_{n=1}^{\infty} \frac{n^n}{n!}$

(28) $\sum_{n=2}^{\infty} \frac{n! 10^n}{n^n}$

(29) $\sum_{n=2}^{\infty} \frac{(2n)!}{5^n (n!)^2}$

(30) $\sum_{n=2}^{\infty} \frac{(n+1)(3n)!}{1 \times 3 \times \dots \times (2n+1) (n!)^3}$

(31) $\sum_{n=2}^{\infty} \frac{(3n)!}{5^n n!}$

(32) $\sum_{n=2}^{\infty} (\sqrt{n+2} - \sqrt{n-2})/n$

(33) $\sum_{n=2}^{\infty} \ln(n^2 + 1) - \ln(n^2)$

(34) $\sum_{n=1}^{\infty} n \sin^3 \frac{1}{n}$

(35) $\sum_{n=2}^{\infty} \frac{e^{\tan^{-1} n}}{n^2}$

(36) $\sum_{n=2}^{\infty} \frac{1}{n^{1+1/n}}$

2. 다음 급수의 수렴 및 발산을 판정하면?

(1) $\sum_{n=2}^{\infty} \frac{\cos n}{n^2}$

(2) $\sum_{n=1}^{\infty} \frac{\sqrt{n} \cos n}{n^3}$

(3) $\sum_{n=1}^{\infty} \frac{\sin n}{n}$

(4) $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$

(5) $\sum_{n=1}^{\infty} \frac{n + \sin n}{n^3}$

(6) $\sum_{n=1}^{\infty} \frac{5 \cos\left(\frac{n\pi}{10}\right)}{\sqrt{3} n^4}$

(7) $\sum_{n=1}^{\infty} \frac{\sin 4n}{4^n}$

(8) $\sum_{n=1}^{\infty} \frac{\cos^2 n}{n(n+1)}$

3. 무한급수의 수렴반경은?

(1) $\sum \frac{n(x-1)^n}{5^n}$

(2) $\sum \frac{(x+1)^n}{3^n + 4^n}$

(3) $\sum \frac{n^n x^n}{n!}$

(4) $\sum \frac{(2n)! x^n}{(n!)^2}$

(5) $\sum \frac{1 \times 3 \times \dots \times (2n+1) (x-1)^n}{(n+3)!}$

(6) $\sum \frac{n! (n+1)! (x-1)^n}{(2n)!}$

(7) $\sum \frac{(n!)^2 (x-1)^n}{(3n)!}$

(8) $\sum \frac{3^n (x+2)^n}{n!}$

4. 무한급수의 수렴구간은?

(1) $\sum \frac{n(x+1)^n}{2^{n+2}}$

(2) $\sum \frac{x^n}{n^3 + 1}$

(3) $\sum \frac{(-1)^{n+1} (x-2)^n}{n+1}$

(4) $\sum \frac{(x+2)^n}{3^n (n^2 + 1)}$

(5) $\sum \frac{3^n (x^2 - 1)^n}{n^2 4^n}$

(6) $\sum \frac{x^n}{n \sqrt{\ln n}}$

(7) $\sum \frac{e^{nx}}{\sqrt{n}}$

(8) $\sum \frac{1}{n^2} \left(\frac{x-2}{x} \right)^n$

<정답표>

- (1) 수렴 (2) 발산 (3) 수렴 (4) 수렴 (5) 수렴 (6) 수렴
(7) 발산 (8) 발산 (9) 발산 (10) 발산 (11) 발산 (12) 수렴
(13) 발산 (14) 발산 (15) 발산 (16) 수렴 (17) 수렴 (18) 수렴
(19) 발산 (20) 발산 (21) 수렴 (22) 발산 (23) 수렴 (24) 수렴
(25) 발산 (26) 발산 (27) 발산 (28) 발산 (29) 수렴 (30) 수렴
(31) 발산 (32) 수렴 (33) 수렴 (34) 수렴 (35) 수렴 (36) 발산
- (1) 수렴 (2) 수렴 (3) 수렴 (4) 수렴 (5) 수렴 (6) 수렴
(7) 수렴 (8) 수렴
- (1) 5 (2) 4 (3) $\frac{1}{e}$ (4) $\frac{1}{4}$ (5) $\frac{1}{2}$ (6) 4
(7) ∞ (8) ∞
- (1) $-3 < x < 1$ (2) $-1 \leq x \leq 1$ (3) $1 < x \leq 3$
(4) $-5 \leq x \leq 1$ (5) $-\sqrt{\frac{7}{3}} \leq x \leq \sqrt{\frac{7}{3}}$ (6) $-1 \leq x < 1$
(7) $x < 0$ (8) $x \geq 1$